

DATABASE SCHEMA

Learning Center Management System

Revised Version — All Agreed Changes Applied

1. People & Accounts

VAITRO — Roles

Field	Type / Constraint	Notes
MAVAITRO	NUMBER PRIMARY KEY	
TENVAITRO	VARCHAR2(50) NOT NULL	'Admin', 'GiaoVien', 'HocVien', 'KeToan'

TAIKHOAN — System Accounts

Field	Type / Constraint	Notes
MATAIKHOAN	NUMBER PRIMARY KEY	
TENTAIKHOAN	VARCHAR2(100) UNIQUE NOT NULL	
MATKHAU	VARCHAR2(255) NOT NULL	BCrypt hashed — never plain text
MAVAITRO	NUMBER FK → VAITRO	
TRANGTHAITK	VARCHAR2(20)	'Hoat dong', 'Vo hieu'
NGAYTAO	TIMESTAMP DEFAULT SYSDATE	NEW — account creation date
LANDN_CUOI	TIMESTAMP	NEW — last login timestamp

★ NGAYTAO and LANDN_CUOI added for security audit trail.

HOCVIEN — Students

Field	Type / Constraint	Notes
MAHOCVIEN	NUMBER PRIMARY KEY	
MATAIKHOAN	NUMBER FK → TAIKHOAN	
HOTEN	VARCHAR2(100) NOT NULL	
NGAYSINH	DATE	
GIOITINH	VARCHAR2(10)	'Nam', 'Nu', 'Khac'
DIACHI	VARCHAR2(255)	
SDT	VARCHAR2(15)	
EMAIL	VARCHAR2(100)	

★ STK and NGANHANG REMOVED — moved to THONGTINNGANHANG for security.

THONGTINNGANHANG — Student Bank Details [NEW TABLE]

Field	Type / Constraint	Notes
MATHONGTIN	NUMBER PRIMARY KEY	

Field	Type / Constraint	Notes
MAHOCVIEN	NUMBER FK → HOCVIEN	
STK	VARCHAR2(30) NOT NULL	Bank account number
NGANHANG	VARCHAR2(100) NOT NULL	'Vietcombank', 'BIDV', 'Techcombank', ...
CHUSOHUU	VARCHAR2(100)	Account owner name
LOAITK	VARCHAR2(10)	'Chinh', 'Phu'
TRANGTHAI	VARCHAR2(20)	'Hoat dong', 'Vo hieu'

★ Access restricted to Accountant role only. Referenced by HOADONHOCPhi for refund bank routing.

NHANVIEN — Staff (includes Teachers)

Field	Type / Constraint	Notes
MANHANVIEN	NUMBER PRIMARY KEY	
HOTEN	VARCHAR2(100) NOT NULL	
GIOITINH	VARCHAR2(10)	
NGAYSINH	DATE	NEW
SDT	VARCHAR2(15)	NEW
EMAIL	VARCHAR2(100)	NEW
CHUCVU	VARCHAR2(50)	'Giao vien', 'Ke toan', 'Quan ly'
MATAIKHOAN	NUMBER FK → TAIKHOAN	

★ Teachers are staff. GIAOVIEN extends NHANVIEN for teacher-specific info.

GIAOVIEN — Teacher Extension

Field	Type / Constraint	Notes
MAGIAOVIEN	NUMBER PRIMARY KEY	
MANHANVIEN	NUMBER FK → NHANVIEN	CHANGED from MATAIKHOAN
MABOMON	VARCHAR2(200)	NEW — specialization
BANGCAP	VARCHAR2(100)	NEW — qualification/degree
TRANGTHAI	VARCHAR2(20)	NEW — 'Dang day', 'Nghiep', 'Da nghi'

★ MABOMON removed — already in NHANVIEN. MATAIKHOAN replaced by MANHANVIEN.

BOMON — Departments

Field	Type / Constraint	Notes
MABOMON	NUMBER PRIMARY KEY	
TENBM	VARCHAR2(100) NOT NULL	

2. Academic

KHOAHOC — Courses

Field	Type / Constraint	Notes
MAKHOAHOC	NUMBER PRIMARY KEY	
TENKH	VARCHAR2(200) NOT NULL	
MOTA	CLOB	
HOCPHITHANG	NUMBER(12,2)	NEW — monthly fee option
HOCPHITOANKH	NUMBER(12,2)	NEW — full course fee option
MABOMON	NUMBER FK → BOMON	
CAPDO	VARCHAR2(50)	'Co ban', 'Trung cap', 'Nang cao'

★ *HOCPHI split into HOCPHITHANG and HOCPHITOANKH for monthly vs full-course payment plans.*

LOPHOC — Classes

Field	Type / Constraint	Notes
MALOPHOC	NUMBER PRIMARY KEY	
TENLOP	VARCHAR2(100) NOT NULL	
NGAYBD	DATE	Start date
NGAYKT	DATE	End date
MAKHOAHOC	NUMBER FK → KHOAHOC	
MAGIAOVIEN	NUMBER FK → GIAOVIEN	
SISO	NUMBER	Max capacity
TANSUAT	VARCHAR2(50)	eg. '3 buoi/tuan'
TRANGTHAI	VARCHAR2(20)	NEW — 'Dang mo', 'Dang hoc', 'Ket thuc', 'Huy'

★ *TRANGTHAI added — controls enrollment availability.*

LICHHOC — Weekly Schedule Templates

Field	Type / Constraint	Notes
MALICH	NUMBER PRIMARY KEY	
MALOP	NUMBER FK → LOPHOC	
GIOBATDAU	VARCHAR2(10)	'08:00', '13:30', ...
GIOKETTHUC	VARCHAR2(10)	
THU	VARCHAR2(10)	'Thu 2', 'Thu 3', ..., 'Chu nhat'

BUOIHOC — Actual Sessions

Field	Type / Constraint	Notes
MABUOIHOC	NUMBER PRIMARY KEY	
MALOPHOC	NUMBER FK → LOPHOC	

Field	Type / Constraint	Notes
MALICH	NUMBER FK → LICHHOC	NEW — links session to schedule template
NGAYHOC	DATE NOT NULL	
GIOHOC	VARCHAR2(10)	
MAPHONGHOC	NUMBER FK → PHONGHOC	
TRANGTHAI	VARCHAR2(20)	'Sap dien ra','Dang hoc','Hoan thanh','Huy'
LOAIBUOI	VARCHAR2(20)	NEW — 'Thuong','Bu','Them'

★ **MALICH** links session to schedule template. NULL allowed for make-up sessions.

★ **LOAIBUOI**='Bu' for make-up sessions. Make-up: **MALICH**=NULL, different **NGAYHOC**/**MAPHONGHOC**.

DIEMDANH — Attendance

Field	Type / Constraint	Notes
MABUOIHOC	NUMBER FK → BUOIHOC [PK]	Composite PK
MAHOCVIEN	NUMBER FK → HOCVIEN [PK]	Composite PK
TRANGTHAI	VARCHAR2(20)	'Co mat','Vang mat','Tre','Phep'

BAIKIEMTRA — Exams / Tests

Field	Type / Constraint	Notes
MABAIKIEMTRA	NUMBER PRIMARY KEY	
TENBKT	VARCHAR2(200) NOT NULL	
NGAYKT	DATE	
THOIGIAN	NUMBER	Duration in minutes
LOAIBKT	VARCHAR2(50)	'Giua ky','Cuoi ky','Thuong xuyen'
MALOP	NUMBER FK → LOPHOC	
DIEMTOIDA	NUMBER	NEW — max score eg. 100
DIEMDAT	NUMBER	NEW — passing score eg. 60

★ **DIEMTOIDA** and **DIEMDAT** added — needed to interpret **DIEMSO** in **THAMGIABKT**.

THAMGIABKT — Exam Participation

Field	Type / Constraint	Notes
MAHOCVIEN	NUMBER FK → HOCVIEN [PK]	Composite PK
MABAIKIEMTRA	NUMBER FK → BAIKIEMTRA [PK]	Composite PK
LANTHI	NUMBER DEFAULT 1 [PK]	Attempt number — part of composite PK
DIEMSO	NUMBER(5,2)	CONSTRAINT: 0 <= DIEMSO <= BAIKIEMTRA.DIEMTOIDA
NHANXET	VARCHAR2(500)	
NGAYTHI	DATE	NEW — exam date

Field	Type / Constraint	Notes
THOIGIANTHI	TIMESTAMP	NEW — exact timestamp of attempt

TAILIEU — Learning Materials

Field	Type / Constraint	Notes
MATAILIEU	NUMBER PRIMARY KEY	
TENTAILIEU	VARCHAR2(200) NOT NULL	
LINK	VARCHAR2(500)	URL or file path
MALOPHOC	NUMBER FK → LOPHOC	

PHONGHOC — Classrooms

Field	Type / Constraint	Notes
MAPHONGHOC	NUMBER PRIMARY KEY	
TENPHONG	VARCHAR2(100) NOT NULL	
SUCHUA	NUMBER	Capacity
TRANGTHAI	VARCHAR2(20)	'San sang', 'Dang dung', 'Bao tri'

3. Enrollment & Class Transfer

DANGKY — Enrollments

Field	Type / Constraint	Notes
MAHOCVIEN	NUMBER FK → HOCVIEN [PK]	Composite PK
MALOPHOC	NUMBER FK → LOPHOC [PK]	Composite PK
NGAYDANGKY	DATE DEFAULT SYSDATE	
TRANGTHAIDKY	VARCHAR2(30)	'Dang hoc', 'Hoan thanh', 'Huy', 'Chuyen lop'
MAHOADON	NUMBER FK → HOADONHOCPhi	NEW — links enrollment to invoice
HINHTHUCTT	VARCHAR2(20)	NEW — '1 thang', 'Toan khoa', 'Theo ky'

★ MAHOADON + HINHTHUCTT added — links enrollment to its invoice and records chosen payment plan.

YEUCAUCHUYENLOP — Class Transfer Requests

Field	Type / Constraint	Notes
MAYEUCAU	NUMBER PRIMARY KEY	
MAHOCVIEN	NUMBER FK → HOCVIEN	FIXED from MAHOC SINH
MALOPCU	NUMBER FK → LOPHOC	Original class
MALOPMOI	NUMBER FK → LOPHOC	Requested new class

Field	Type / Constraint	Notes
LYDO	VARCHAR2 (500)	
NGAYYEUCAU	DATE DEFAULT SYSDATE	
TRANGTHAI	VARCHAR2 (20)	'Cho duyet','Chap thuan','Tu choi'
MANHANVIEN	NUMBER FK → NHANVIEN	NEW – staff who processed
NGAYXULY	DATE	NEW – date of decision
GHICHU	VARCHAR2 (500)	NEW – admin notes

★ *MAHOCSINH corrected to MAHOCVIEN for consistency across all tables.*

★ *Approval flow: YEUCAUCHUYENLOP approved -> DANGKY updated (old row: 'Chuyen lop', new row inserted).*

4. Financial

GIAMGIA — Discount Programs

Field	Type / Constraint	Notes
MAGIAMGIA	NUMBER PRIMARY KEY	
TENCHUONGTRINH	VARCHAR2 (200) NOT NULL	
PHANTRAM	NUMBER (5,2)	Discount % eg. 10.00
HANDUNG	DATE	Expiry date
SOLANSUDUNG	NUMBER	Max usage count

HOCBONG — Scholarships

Field	Type / Constraint	Notes
MAHOCBONG	NUMBER PRIMARY KEY	
MAHOCVIEN	NUMBER FK → HOCVIEN	
GIATRI	NUMBER (12,2)	Value in VND
NGAYCAP	DATE	
TRANGTHAIHB	VARCHAR2 (20)	'Con hieu luc','Het han','Da su dung'

HOADONHOCPhi — Invoices

Field	Type / Constraint	Notes
MAHOADON	NUMBER PRIMARY KEY	
MAHOCVIEN	NUMBER FK → HOCVIEN	
MAGIAMGIA	NUMBER FK → GIAMGIA	Nullable – max one discount
MAHOCBONG	NUMBER FK → HOCBONG	Nullable
MALOP	NUMBER FK → LOPHOC	
NGAYLAP	DATE DEFAULT SYSDATE	
TONGTIEN	NUMBER (12,2)	Negative for outgoing corrections

Field	Type / Constraint	Notes
TRANGTHAIHD	VARCHAR2 (20)	'Cho thanh toan', 'Da thanh toan', 'Huy'
LOAIHD	VARCHAR2 (50)	'Dong tien hoc phi', 'Hoc bong', 'Dieu chinh', 'Phi tre han'
HINHTHUC	VARCHAR2 (50)	'Tien mat', 'Chuyen khoan', 'Mock card'
HINHTHUCTT	VARCHAR2 (20)	NEW — '1 thang', 'Toan khoa', 'Theo ky'
THANGTT	NUMBER (2)	NEW — month 1-12; NULL if full course
NAMTT	NUMBER (4)	NEW — year eg. 2026; NULL if full course
MAHOADON_GOC	NUMBER FK → HOADONHOCPhi	NEW — self-ref for corrections; NULL for normal
MATHONGTIN	NUMBER FK → THONGTINNGANHANG	NEW — bank account for refunds
GHICHU	VARCHAR2 (500)	NEW — notes/reason

★ **LOAIHD** no longer includes 'Hoan tra' — refunds handled by separate **HOANTRA** table.

★ **HINHTHUCTT** + **THANGTT** + **NAMTT** support monthly and full-course payment tracking.

HOANTRA — Refund Records [NEW TABLE]

Field	Type / Constraint	Notes
MAHOANTRA	NUMBER PRIMARY KEY	
MAHOADON	NUMBER FK → HOADONHOCPhi	Original invoice being refunded
MAHOCVIEN	NUMBER FK → HOCVIEN	
MATHONGTIN	NUMBER FK → THONGTINNGANHANG	Bank account to receive refund
SOTIEN	NUMBER (12, 2) NOT NULL	Refund amount
LYDO	VARCHAR2 (500)	Reason for refund
NGAYYEUCAU	DATE DEFAULT SYSDATE	
NGAYHOANTRA	DATE	Actual processed date
TRANGTHAI	VARCHAR2 (20)	'Cho duyet', 'Da hoan', 'Tu choi'
MANHANVIEN	NUMBER FK → NHANVIEN	Accountant who processed
NGAYTAO	TIMESTAMP DEFAULT SYSDATE	Audit — creation timestamp
NGUOITAO	NUMBER FK → TAIKHOAN	Audit — who created the request

★ Separate from **HOADONHOCPhi** for cleaner logic. Links to original invoice via **MAHOADON**.

5. System & Scheduling

NHATKYHETHONG — System Logs

Field	Type / Constraint	Notes
MANHATKY	NUMBER PRIMARY KEY	
THOIGIAN	TIMESTAMP DEFAULT SYSDATE	CHANGED from NGAY (date-only) to TIMESTAMP
MATAIKHOAN	NUMBER FK → TAIKHOAN	
LOAITHAOTAC	VARCHAR2 (20)	NEW – 'INSERT', 'UPDATE', 'DELETE', 'LOGIN', 'LOGOUT'
TENBANGGHI	VARCHAR2 (50)	NEW – affected table eg. 'HOCVIEN'
MABANGHI	NUMBER	NEW – affected record ID
MOTA	VARCHAR2 (500)	NEW – description of change
DIACHIIP	VARCHAR2 (45)	NEW – IP address (supports IPv6)

★ NGAY and THAOTAC replaced with structured fields for meaningful audit queries.

NGAYNGHILE — Public Holidays

Field	Type / Constraint	Notes
MANGAYNGHI	NUMBER PRIMARY KEY	
NGAYBATDAU	DATE NOT NULL	
NGAYKETTHUC	DATE NOT NULL	

6. Change Summary

All changes applied per team discussion. NEW = added field/table. CHANGED = modified. FIXED = corrected error.

Table	Change
HOCVIEN	Removed STK, NGANHANG
THONGTINNGANHANG	NEW TABLE – isolated bank details
TAIKHOAN	Added NGAYTAO, LANDN_CUOI
NHANVIEN	Added NGAYSINH, SDT, EMAIL
GIAOVIEN	MATAIKHOAN → MANHANVIEN; added CHUYENMON, BANGCAP, TRANGTHAI; removed MABOMON
KHOAHOC	HOCPhi split into HOCPHITHANG + HOCPHITOANKH
LOPHOC	Added TRANGTHAI
BUOIHOC	Added MALICH (FK→LICHHOC), LOAIBUOI
BAIKIEMTRA	Added DIEMTOIDA, DIEMDAT
THAMGIABKT	Added NGAYTHI, THOIGIAN THI
DANGKY	Added MAHOADON, HINH THUC TT
YEUCAUCHUYENLOP	MAHOC SINH → MAHOC VIEN; added MANHANVIEN, NGAYXULY, GHICHU
HOADONHOCPhi	Added HINH THUC TT, THANG TT, NAM TT, MAHOADON_GOC, MATHONG TIN, GHICHU; LOAI HD updated
HOANTRA	NEW TABLE – dedicated refund tracking
NHATKYHETHONG	Restructured: NGAY → THOIGIAN, THAOTAC → 5 structured fields

Table	Change
PHONGHOC	No change (small facility – room type not needed)
GIAMGIA	No change (one discount per invoice accepted)
KHOAHOC prerequisites	Deferred – free enrollment policy chosen